

We have that

$$S_{n+1}^2 = \frac{\sum_{i=1}^{n+1} (x_i - \bar{x}_{n+1})^2}{(n+1) - 1}$$

by (a)

$$= \frac{\sum_{i=1}^{n+1} (x_i - \bar{x}_{n+1})^2}{n}$$

$$\begin{aligned} \text{So } n S_{n+1}^2 &= \sum_{i=1}^{n+1} (x_i - \bar{x}_{n+1})^2 = \sum_{i=1}^{n+1} \left(x_i - \frac{n\bar{x}_n + x_{n+1}}{n+1} \right)^2 \\ &= \sum_{i=1}^{n+1} \left(\frac{(n+1)x_i - n\bar{x}_n - x_{n+1}}{(n+1)} \right)^2 \\ &= \sum_{i=1}^{n+1} \left(\frac{(n+1)x_i - (n+1)\bar{x}_n + \bar{x}_n - x_{n+1}}{(n+1)} \right)^2 \end{aligned}$$

$$\begin{aligned} \Rightarrow n S_{n+1}^2 &= \sum_{i=1}^{n+1} \left((x_i - \bar{x}_n) - \left(\frac{x_{n+1} - \bar{x}_n}{n+1} \right) \right)^2 \\ &= \sum_{i=1}^{n+1} \left((x_i - \bar{x}_n)^2 - 2(x_i - \bar{x}_n) \left(\frac{x_{n+1} - \bar{x}_n}{n+1} \right) + \left(\frac{x_{n+1} - \bar{x}_n}{n+1} \right)^2 \right) \\ &= \sum_{i=1}^{n+1} (x_i - \bar{x}_n)^2 - 2 \left(\frac{x_{n+1} - \bar{x}_n}{n+1} \right) \sum_{i=1}^{n+1} (x_i - \bar{x}_n) + (n+1) \left(\frac{x_{n+1} - \bar{x}_n}{n+1} \right)^2 \end{aligned}$$

$$\Rightarrow \text{Now notice } \sum_{i=1}^{n+1} (x_i - \bar{x}_n)^2 = \sum_{i=1}^n (x_i - \bar{x}_n)^2 + (x_{n+1} - \bar{x}_n)^2$$

$$= (n-1) S_n^2 + (x_{n+1} - \bar{x}_n)^2$$

$$\text{and } \sum_{i=1}^{n+1} (x_i - \bar{x}_n) = \sum_{i=1}^n (x_i - \bar{x}_n) + (x_{n+1} - \bar{x}_n) = 0 + (x_{n+1} - \bar{x}_n)$$

$$\begin{aligned} \text{So } n S_{n+1}^2 &= (n-1) S_n^2 + (x_{n+1} - \bar{x}_n)^2 - 2 \left(\frac{x_{n+1} - \bar{x}_n}{n+1} \right) (x_{n+1} - \bar{x}_n) \\ &\quad + (n+1) \cdot \left(\frac{x_{n+1} - \bar{x}_n}{n+1} \right)^2 \\ &= (n-1) S_n^2 + 1 \cdot (x_{n+1} - \bar{x}_n)^2 - \frac{2}{n+1} (x_{n+1} - \bar{x}_n)^2 + \frac{1}{(n+1)} (x_{n+1} - \bar{x}_n)^2 \\ &= (n-1) S_n^2 + \frac{n}{n+1} (x_{n+1} - \bar{x}_n)^2 \\ &\text{as required} \end{aligned}$$